



Stages of the data life cycle

Here's a question for you.

When you think about a life cycle, what's the first thing that comes to mind?

Now I'm not a mind reader, but I know whatever you're thinking is right.

There's actually no wrong answer because everything has a life cycle.

One of the most well known examples of a life cycle is a butterfly.

Butterflies begin as eggs, hatch into caterpillars and then become a **chrysalis**.

That's where the real magic happens.

Data has a life cycle of its own too.

In this video, we're going to talk about each of the stages in that life cycle

to help you understand the individual phases data goes through.

The life cycle of data is **plan**, **capture**, **manage**,
analyze, **archive** and **destroy**.

Let's start with the **first phase**, **planning**.

This actually happens well **before** starting an analysis project.

During planning, **a business decides what kind of data it needs**,
how it will be managed throughout its life cycle,
who will be responsible for it, **and the optimal outcomes**.

For example, let's say an electricity provider wanted to gain insights
into how to save people energy.

In the planning phase, they might decide to capture information on how much
electricity its customers use each year,

what types of buildings are being powered, and what types of devices are being powered inside of them. The electricity company would also decide which team members will be responsible for collecting, storing, and sharing that data. All of this happens during planning, and it helps set up the rest of the project. The **next phase** is when you **capture data**.

This is where data is collected from a variety of different sources and brought into the organization.

With so much data being created everyday, the ways to collect it are truly endless.

One common method is getting data from outside resources. For example, if you were doing data analysis on weather patterns, you'd probably get data from a publicly available dataset like the National Climatic Data Center.

Another way to get data is from a company's own documents and files, which are usually stored inside a database. While we've mentioned databases before, we haven't gone into too much detail about what they are.

A database is a collection of data stored in a computer system.

In the case of our electricity provider, the business would probably measure data usage among its customers within a database that it owns. As a quick note, **when you maintain a database of customer information, ensuring data integrity, credibility, and privacy are all important concerns.**

You'll learn a lot more about that later on.

Now that we've captured our data, we'll move on to the **next phase** of the data life cycle, **manage**.

Here we're talking about how we care for our data, how and where it's stored, the tools used to keep it safe and secure, and

the actions taken to make sure that it's maintained properly.

This phase is very important to data cleansing, which we'll cover later on.

Next it's time to analyze your data. This is where data analysts really shine.

In this phase, the data is used to solve problems, make great decisions, and support business goals.

For example, one of our electricity company's goals might be to find ways to help customers save energy.

Moving along the data life cycle now evolves to the archive phase.

Archiving means storing data in a place where it's still available, but may not be used again.

During analysis, analysts handle huge amounts of data.

Can you imagine if we had to sort through all of the available data that's out there, even if it was no longer useful and relevant to our work?

It makes way more sense to archive it than to keep it around.

And finally, the last step of the data life cycle, the destroy phase.

Yes, it sounds sad, but when you destroy data, it won't hurt a bit.

So let's get back to our electricity provider example.

They would have data stored on multiple hard drives.

To destroy it, the company would use a secure data erasure software.

If there were any paper files, they would be shredded too.

This is important for protecting a company's private information, as well as private data about its customers.

And there you have it, the data life cycle.

And now that you understand the different phases data goes through during its life cycle, you can better understand how to approach the data analysis process, which we'll talk about soon.